# AAE1001 - Introduction to Artificial Intelligence and Data Analytics in Aerospace and Aviation Engineering

## Week 6-11: Group Project - Design of Path Planning Algorithm for Aircraft Operation

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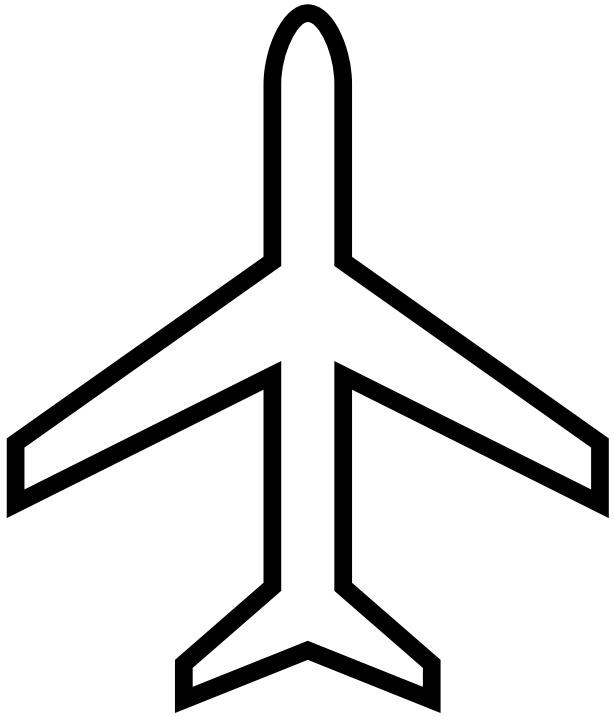
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# Week 6 Content



1. Introduction to Path Planning
2. Introduction to GitHub (Background)
3. Introduction to GitHub Operations
4. Software Installation and setup Guide

# Introduction to Path Planning

# Planning in AI: Go and Flight Routes

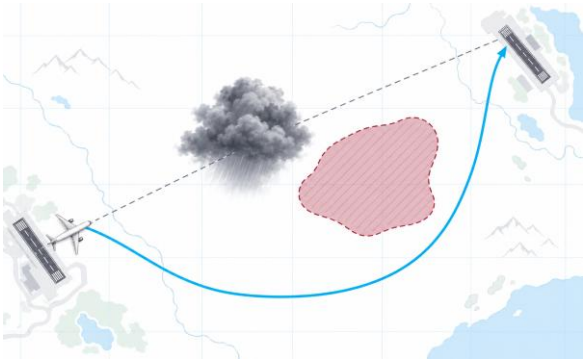
## Why look ahead before choosing a move or a route?

### Go: anticipate the next moves




AlphaGo combines learned models with search over moves and opponent replies.

### Flight: compare possible routes



A planner compares routes to a destination, respecting hazards and operating costs.

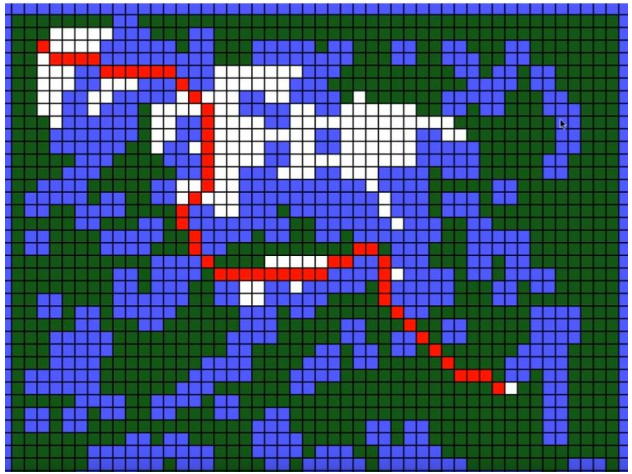
**Planning is part of AI: choose actions toward a goal by looking ahead.**

Why learn it? Turn predictions into decisions; compare feasible routes and costs.

# What is Path Planning?

**How to go from A to B considering factors!**

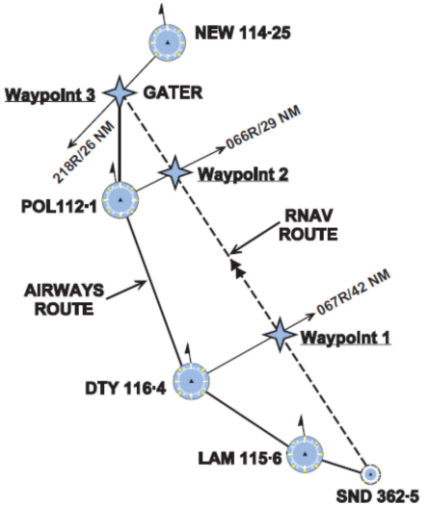
- **Path planning** (also known as the **navigation problem**) is computational problem to find a sequence of valid configurations that moves the object from the source to destination. The term is used in **aviation**, **robotics** and **computer games**.



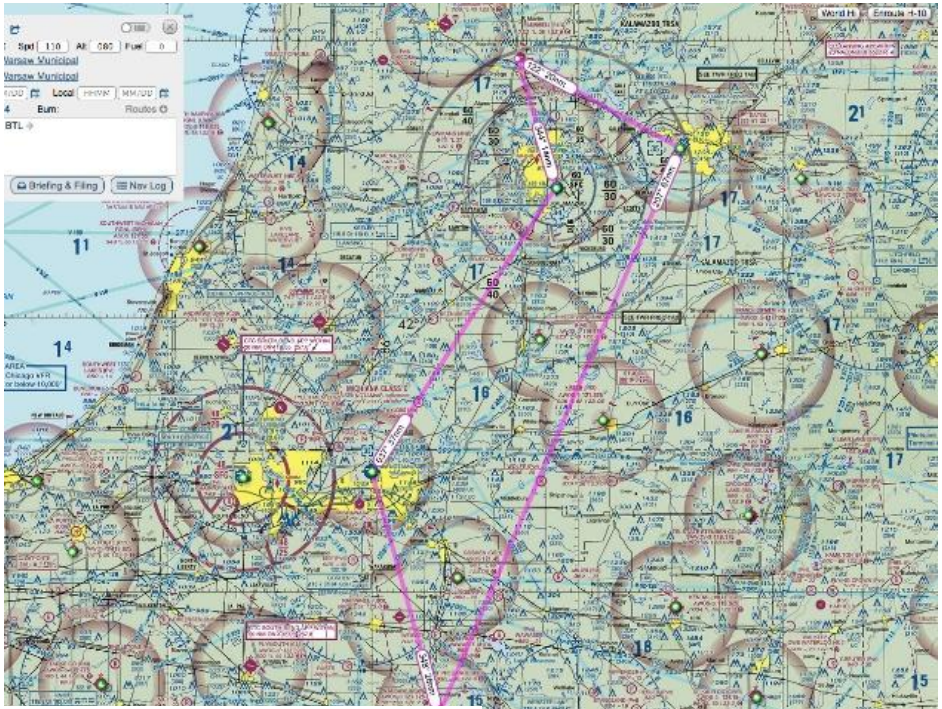


# How is Path Planning important to Aviation Engineering?

- Private pilots do the path plan before the flight to make sure the navigation aid is available





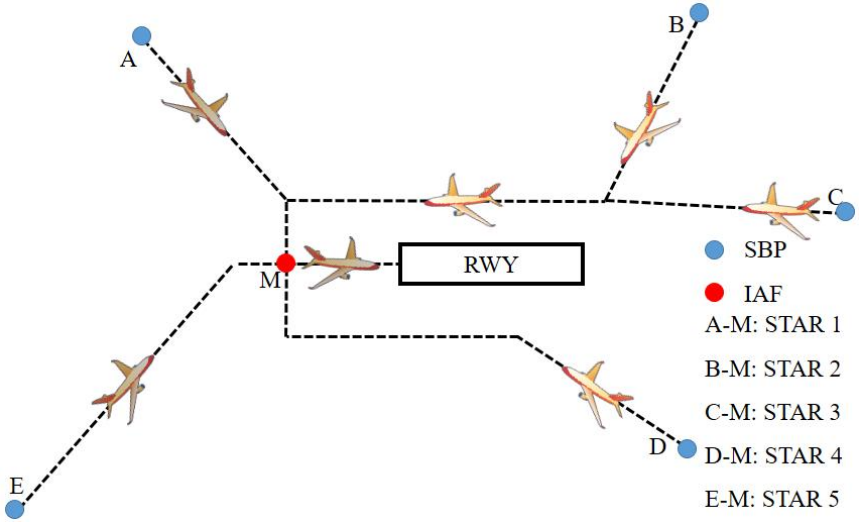



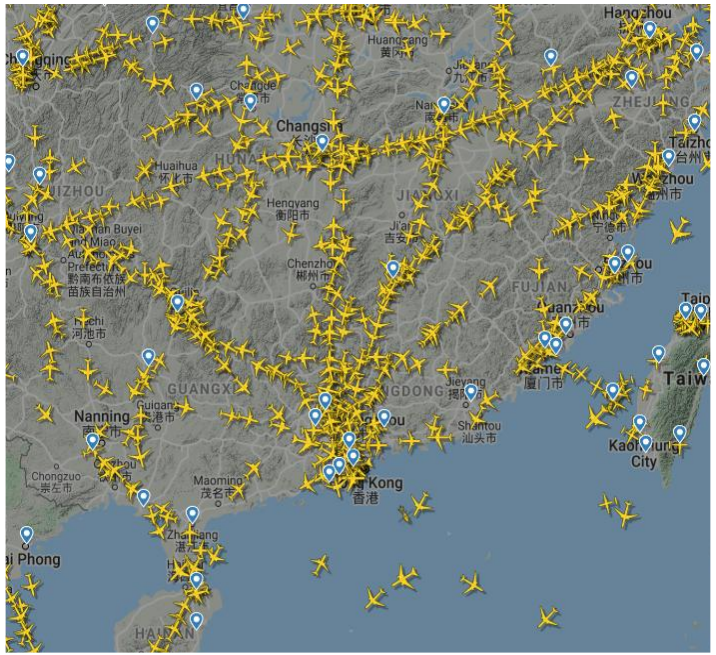



Objective: Safe and Best Sight Seeing

# How is Path Planning important to Aviation Engineering?

- For ATC near airports, collaborative path planning is required to make the best use of the crowded airspace



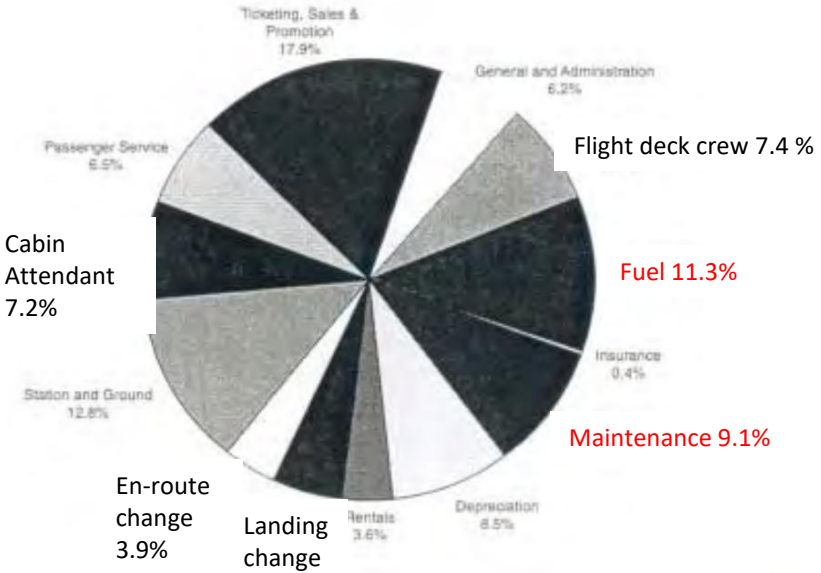

Objective: Safe and least delay

# How is Path Planning important to Aviation Engineering?

- Commercial pilot follow the path that plan based on different cost index designed by airlines.

Objective: Safe and Minimum Cost

**Figure 2. Distribution of operating costs**




**Source : IATA**

https://ansperformance.eu/library/airbus-cost-index.pdf

## 2.1 Trip cost

Without having to resort to complicated mathematics we can readily appreciate that the total cost of a specific trip is the sum of fixed and variable costs :

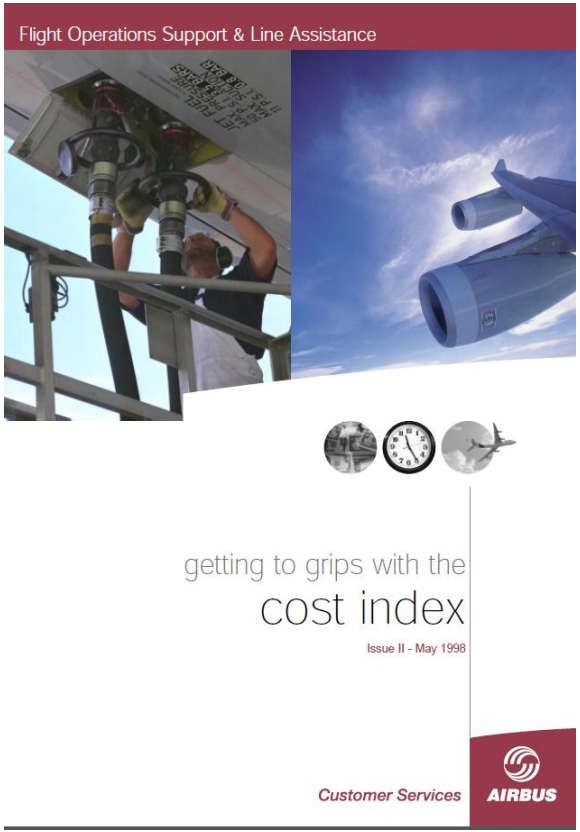
$$C = C_F \times \Delta F + C_T \times \Delta T + C_c$$

with $C_F$ = cost of fuel per kg
$C_T$ = time-related cost per minute of flight
$C_c$ = fixed costs independent of time
$\Delta F$ = trip fuel
$\Delta T$ = trip time

In order to minimize C or the total trip cost we therefore need to minimize the variable cost :

$$C_F \times \Delta F + C_T \times \Delta T$$

# Cost-Index Published by Aircraft Manufacturer




AIRBUS

**3.1 A300/A310 Family**

Considering, with good approximation, that the following range of time-related costs cover the maintenance cost difference between A300 and A310 as well as the cabin crew contingent (plus or minus two) difference, the following cost brackets result :

| | | |
|---|---|---|
| | 6 < Hourly maintenance cost | < 12 (US$/min) |
| + | 7 < Crew cost | < 14 (US$/min) |
| | 13 < Time-related cost | < 26 (US$/min) |

NB : Crew composition = 2 cockpit crews + 8 (± 2) cabin crews.

In turn, the following cost index tables reflect these cost ranges for the A300 and for the A310.

**Table 1. A300/A310 cost index**

(kg/min)
(Honeywell FMS)

| TIME COST (US$/min) / FUEL COST (US$/USG) | LOW < 15 | MEDIUM 15 < to < 20 | HIGH > 20 |
|---|---|---|---|
| LOW < 0.7 | **65** | **85** | **100** |
| MEDIUM 0.7 < < 0.9 | **50** | **65** | **80** |
| HIGH > 0.9 | **40** | **55** | **65** |

https://ansperformance.eu/library/airbus-cost-index.pdf

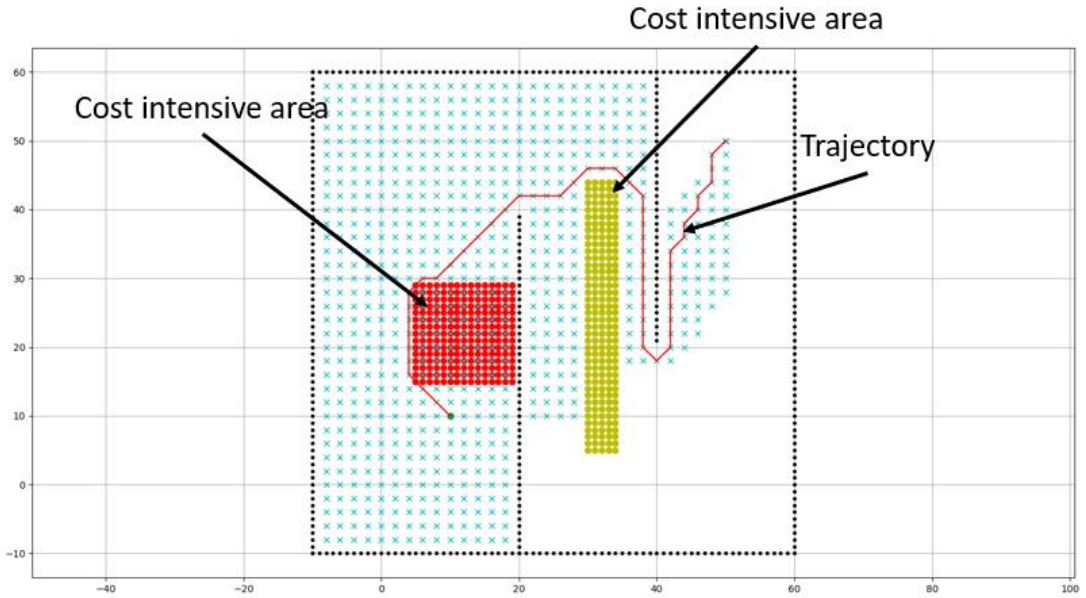
# Path Planning

- Optimization Problem:
- To optimize a path that fulfilling all the constrains and by a set of certain criteria.
- Goal of this project, ***to select the best aircraft models with an optimized route that minimized the cost of the aircraft operation under given scenario.***
- ***Design the cost of the aircraft operation***
- ***Design an aircraft model (virtually) with different cost coefficients to fly safe and cheapest.***
- ***Design the path planning algorithm considering 3D, 2D + time, scenarios.***

# Expected Outcome:
# Every Group have different scenarios

We are airline. We wish to find  a route with minimized cost.

What tasks we do?



# Aircraft Operation in Flight Control System

Path Planning

Desired Waypoints, Destination and More

Database

Desired Speed, Height and Heading

Desired Pitch, Yaw and Roll Angles

Change of Elevator, Rudder and Ailerons Positions

Change of Pitch, Yaw and Roll Angles

Airline Plan → Pilot → FMS → Autopilot → FBW → Flight Mechanics → Integration → Position, Velocity and Rotation of the Aircraft

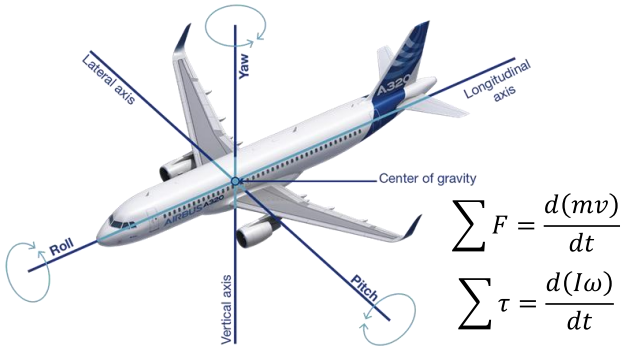
Gyroscopes

Air Data Sensors/Computer

Magnetometer/ AHRS/IRS/GPS

GPS/VOR/DME

VHF/HF by ATC ADS-B/TCAS




$$\sum F = \frac{d(mv)}{dt}$$

$$\sum \tau = \frac{d(I\omega)}{dt}$$

| Climb: | | | Cruise: | | | Descent: | | |
|---|---|---|---|---|---|---|---|---|
| Thrust | > | Drag | Thrust | = | Drag | Thrust | < | Drag |
| Lift | > | Weight | Lift | = | Weight | Lift | < | Weight |

# How is the Path Planning Project related to the AE programme study?

- Mathematics & Physics
- Computer Science
- Aeronautical and Aviation

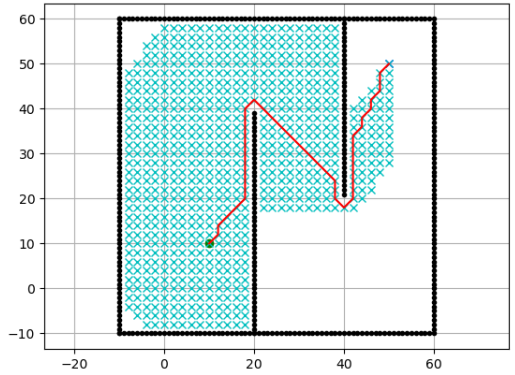
- The path should be planned considering the **practical limitation (aviation system)** of the aircraft

| **Year 1 (31 + 2 training credits)** | |
|---|---|
| **Semester 1 (18 + 1 training credits)** | **Semester 2 (13 + 1 training credits)** |
| AAE1001 Introduction to Artificial Intelligence and Data Analytics in Aerospace and Aviation Engineering (GUR-AIDA) (*2 credits*) | AAE1002 Innovation and Entrepreneurship in Green Aviation and Space Economy (GUR-IE) (*1 credit*) |
| AMA1110 Basic Mathematics I | APSS1L01 Tomorrow's Leaders |
| ENG1004 Engineering Professional in Society I (1 credits) | ENG2002 Computer Programming |
| CAR I ^ | AMA1120 Basic Mathematics II (Faculty Elective II) |
| AP10001 Introduction to Physics (Free elective (1)) | LCR II English Language Subject |
| AP10005 Physics I (Faculty Elective I) | |
| LCR I AI-as-Tool for Language Learning (AITLL) | |
| Healthy Lifestyle (non-credit bearing) ^ | |
| AAE2103/IC2117 – Applied Engineering Fundamentals (2 training credits) | |
| **Year 2 (36 + 4 training credits)** | |
| **Semester 1 (19 + 2 training credits)** | **Semester 2 (17 + 2 training credits)** |
| AAE1003 Introduction to Generative Artificial Intelligence and Digital Legacy in Aviation (1 credit) | AAE2005 Electrics and Electronic for Aeronautical Engineering |
| AAE2003 Introduction to Aircraft Systems | AAE2XXX* Software Applications in Aeronautical Engineering |
| AAE2004 Introduction to Aviation System and Air Transport Regulation | AMA2112 Mathematics II |
| AMA2111 Mathematics I | ENG3006 Engineering Professional in Society II (2 credits) |
| AP10006 Physics II | LCR III Chinese Language Subject |
| ENG2001 Fundamentals of Materials Science and Engineering | Service Learning ^ |
| CAR II^ | AAE3107 / IC385 Fundamentals of Aircraft Fabrication Techniques (2 training credits) |
| AAE3106 / IC384 Fundamentals of Aircraft Engineering Practice (2 training credits) | |
| **Year 3 (21 + 4 training credits)** | |
| **Semester 1 (9 + 2.5 training credits)** | **Semester 2 (12 + 1.5 training credits)** |
| AAE3XXX* Mechanics for Aeronautical Engineering | AAE3001 Fundamentals of Aerodynamics |
| AAE3004 Dynamical Systems and Control | AAE3002 Aircraft Structures and Materials |
| AAE3008 Fundamental Thermal-fluid Science | AAE3003 Aircraft Propulsion Systems |
| AAE3108 / IC386 Aircraft Inspection Practice (1 training credit) | AAE4006 Flight Mechanics and Control Systems |
| AAE3105 / IC389 Aviation Engineering Integrated Project (3 training credits) | |

# What you are expected to learn?

Academic level of algorithm designs

- Design of a path planning algorithm and aircraft model cost function
  - 2D path planning for simplicity




Make use of the **open-resource** to work on coding-project **remotely**.

- Programming and coding
  - Python
- Online coding collaboration
  - GitHub

python™

GitHub

AtsushiSakai / PythonRobotics

# Assessments

- 25%: Presentation and Q&A
- 25%: Report (GitHub repository page by group and individual reflective essay)
- 10%: Performance/participation in in-class activities (Confidential peer evaluation)

* Assessment: 35% Laboratory = 25% report + 10% peer evaluation

# In this project, students will be acted as

- Technical Lead/Members (MUST be the one who is interested at coding)
  - Develop the math behind the compulsory tasks.
  - Write the codes (Compulsory Tasks 1, 2 and 3)
  - Write the codes (Additional Tasks)

- Project Lead/Members (manage the project report and presentation [slide])
  - Study the “numbers” related to aviation context.
  - Prepare report using GitHub Readme format.
  - Prepare PowerPoint slide and the presentation.

# Deadline to submit the work

- Codes uploads to GitHub project created yourself
- Project report written in GitHub readme format.
- Peer review assessment
- On-site presentation (15 minutes).
  - Everyone has to present.
  - Must have a slide
- Report Deadline 23:50pm on 28 Nov (Sat).

# Project GitHub Homepage

https://github.com/RNS-Lab-PolyU/PolyU_AAE1001_Github_Project_2026/

